Sequence 3

STRAIGHT LINES IN THE PLANE

1–16 October 2026

I Direction vector and Cartesian equation

Definition

A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs (AB). Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

Example 3.1

For $A(-2;4)$ and $B(6;2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4;-1)$ directs (AB).

Solution:

Establishing an equation with the determinant

The line through $A(x_A;y_A)$ with direction vector $\vec{u}(\alpha;\beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

Proof to write.

Solution:

Watch out. The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

II Determining the equation of a line

Method

From a point and a direction vector: choose a, b then replace x, y by the coordinates of the point to find c. From two distinct points: first calculate a direction vector.

Example 3.2

Determine a Cartesian equation of the line through $A(2;-1)$ with direction vector $\vec{u}(3;2)$.

Solution:

Example 3.3

Determine a Cartesian equation of (AB) for $A(3;3)$ and $B(2;-2)$.

Solution:

Example 3.4

Does the point $P(1;2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2;1)$ belong to it?

Solution:

III Slope-intercept equation, slope and affine function

Definitions

A non-vertical line has a unique slope-intercept equation $y = mx + p$. m is the slope, p the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1; m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of x. A horizontal line has an equation $y = k$.

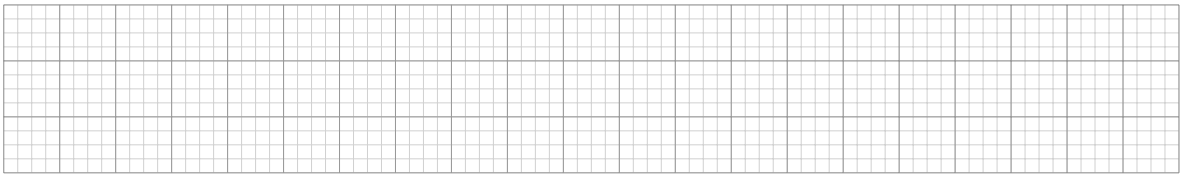
Calculating a slope

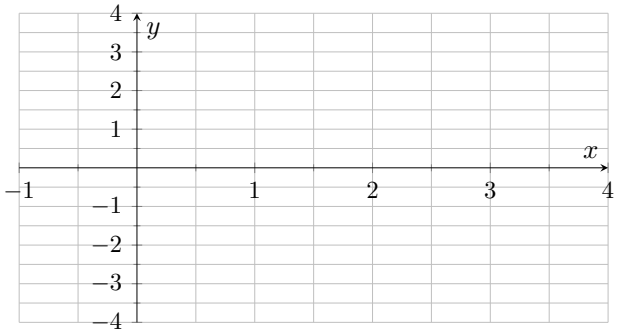
If $x_A \neq x_B$, the slope of (AB) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find p, we then use $y_A = mx_A + p$.

Example 3.5

Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.

Solution:

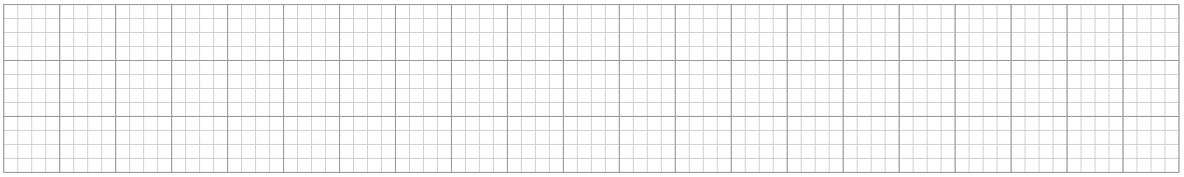




Example 3.6

Find the affine function f such that $f(0) = 3$ and $f(-1) = 5$, then the affine function g whose graph passes through $C(-1; 2)$ and $E(5; -2)$.

Solution:



IV Relative positions of two lines

Properties

Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

Example 3.7

Determine an equation of d through $A(2;3)$ and parallel to $d' : x - 3y - 3 = 0$.

Solution:

Example 3.8

Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

Solution:

V Intersection: solving by substitution

Definition

Solving a system of two equations in x, y means finding all the pairs $(x; y)$ satisfying both equations simultaneously. For two equations of lines: an intersection gives one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

Substitution

Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

Example 3.9

Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

Solution:

Example 3.10

Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

Solution:

VI Linear combination and Python

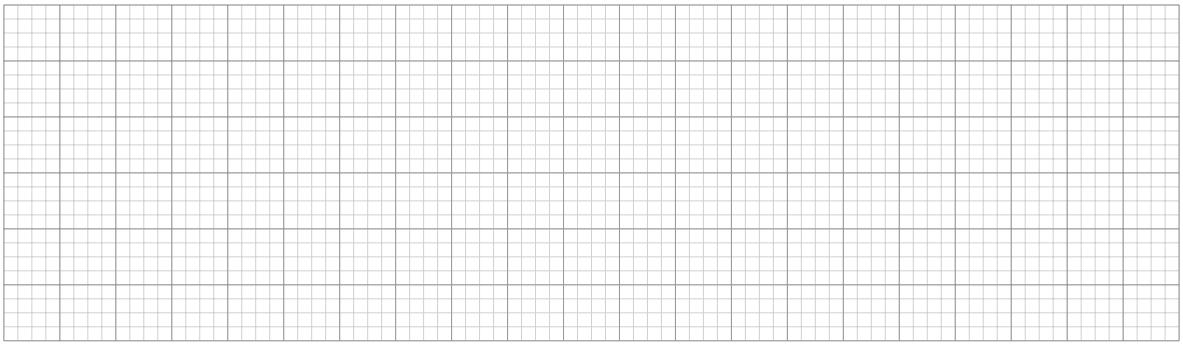
Linear combination

Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

Example 3.11

Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems of the previous example by linear combination.

Solution:



```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

Example 3.12

Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

Solution:

